SANKALP | A PLATFORM FOR CHANGE

Anticipating Diabetic Crisis 120 Minutes Before the First Symptom.

AURA-Twin

AI-Powered **Metabolic Digital Twin** & **Clinical Decision Intelligence** for Type 2 Diabetes Care

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From Continuous Sensor Stream to Bedside Intervention in Real Time.

The Problem

THE HUMAN COST

101M

Indians living with **diabetes**; 136M more are pre-diabetic (ICMR-INDIAB, The Lancet 2023)

1.2M

premature deaths every year in India: one death every **26 seconds**

6.3-8.2 yrs

of **life lost** to glycemic variability and complications (Lancet Global Health)

180 mg/dL

Postprandial spikes drive vascular strain long before HbA1c reflects it

THE ECONOMIC BLEED

$85B+

Annual economic loss from diabetes care, lost workforce and emergency admissions

15-20%

of **hospital inpatient beds** occupied by avoidable diabetes complications (DKA, HHS, nephropathy)

400M

workdays lost each year to diabetic fatigue, clinic visits and hospitalization

THE SYSTEM FAILURE: EPISODIC CARE FOR A REAL-TIME CRISIS

90 days

between HbA1c checks. At 9.5%, that is three months of irreversible vascular damage.

96-288

CGM readings per patient per day, with no clinician time to review them.

Paper-speed

checkups fighting a real-time metabolic crisis.

The Flaw in Current Systems

Retrospective, never predictive

HbA1c and fasting labs show **last month**. They cannot forecast the spike due two hours from now.

The alarm trap

Static **180 mg/dL** alarms fire late and exceed **35% false alarms**, so alerts get silenced.

Black-box readings

A high number with **no cause**: dawn phenomenon, meal glycemic index, missed basal insulin, or renal lag?

Siloed multimodal data

EHR notes, CGM streams, CT/MRI scans and genomics live in **separate systems**.

Dimension	Legacy EHR and standard CGM	AURA-Twin
Response window	Reactive: alerts only after the spike occurs	**Proactive 120-minute** early warning
False alarm rate	High, above **35%**, driving nurse alarm fatigue	Tunable **3.9% to 7.4%** FPR through threshold modes
Explanation	None: raw numbers, black box	**TreeSHAP** names the exact physiological drivers
Scenario testing	Trial and error on the live patient	**In silico** walking, carb-cut and food-sequencing tests
Data silos	Disconnected PDFs, sensors and lab sheets	Unified **FHIR R4** canvas with 3D anatomy and genomics

Our Solution

1	**Clinical Cohort and Temporal Features**	Trained on the **Shanghai T2DM** cohort: **100 patients**, FreeStyle Libre CGM, **15,467** test vectors, with multi-lag velocities, meal timing, circadian transforms and renal biomarkers.
2	**Dual-Head AI Engine**	Head 1: calibrated **XGBoost** flags a **180 mg/dL** spike within **120 minutes**. Head 2: a regressor forecasts glucose at **8 horizons**, +15 to +120 minutes.
3	**Clinician-Tuned Operating Curves**	Switch between outpatient screening (**76.2% recall**) and ward telemetry (**96.1% specificity**, **67.2% precision**, under **4%** false alarms).
4	**TreeSHAP Explainability**	Every prediction ranks its **top drivers**: circadian cosine hour, glucose velocity, minutes since meal, BMI, HbA1c and eGFR.
5	**In Silico Scenario Lab**	Test a **30-minute walk**, a **25 g carb cut** or food sequencing on the twin first, bounded by Clarke Error Grid zones (**99.73%** A+B).
6	**Sourced Evidence Assistant**	**Zero-hallucination** answers: rule-grounded, exact lab timestamps, live glucose charts and NIDDK clinical evidence citations.

Key Features and Engineering Highlights

Real-Time CGM Playback

Timeline scrubbing across **100 patient profiles** with strict zero-leakage temporal features.

Multi-Horizon Forecasting

Future glucose curves: **+15m MAE 4.82** and **+30m MAE 9.43 mg/dL**.

3D Anatomy and CT-to-Mesh

WebGL organ layers (pancreas, liver, kidneys, vasculature) via **Marching Cubes**.

Patient Genomics Panel

Synthetic VCF screened against **36** diabetes risk loci, incl. TCF7L2, KCNJ11, PPARG.

HL7 FHIR R4 Export

One-click bundles of **Observations, RiskAssessments** and **CarePlans**.

Role-Gated Workflow

Drafts, conflict resolution, **version locks** and a full cross-service audit trail.

ENGINEERING HIGHLIGHTS

0.0907

Dual isotonic calibration
Brier score cut in half from 0.1726 (heuristic).

99.73%

Clarke Error Grid safety
78.58% Zone A + 21.15% Zone B over 15,467 vectors.

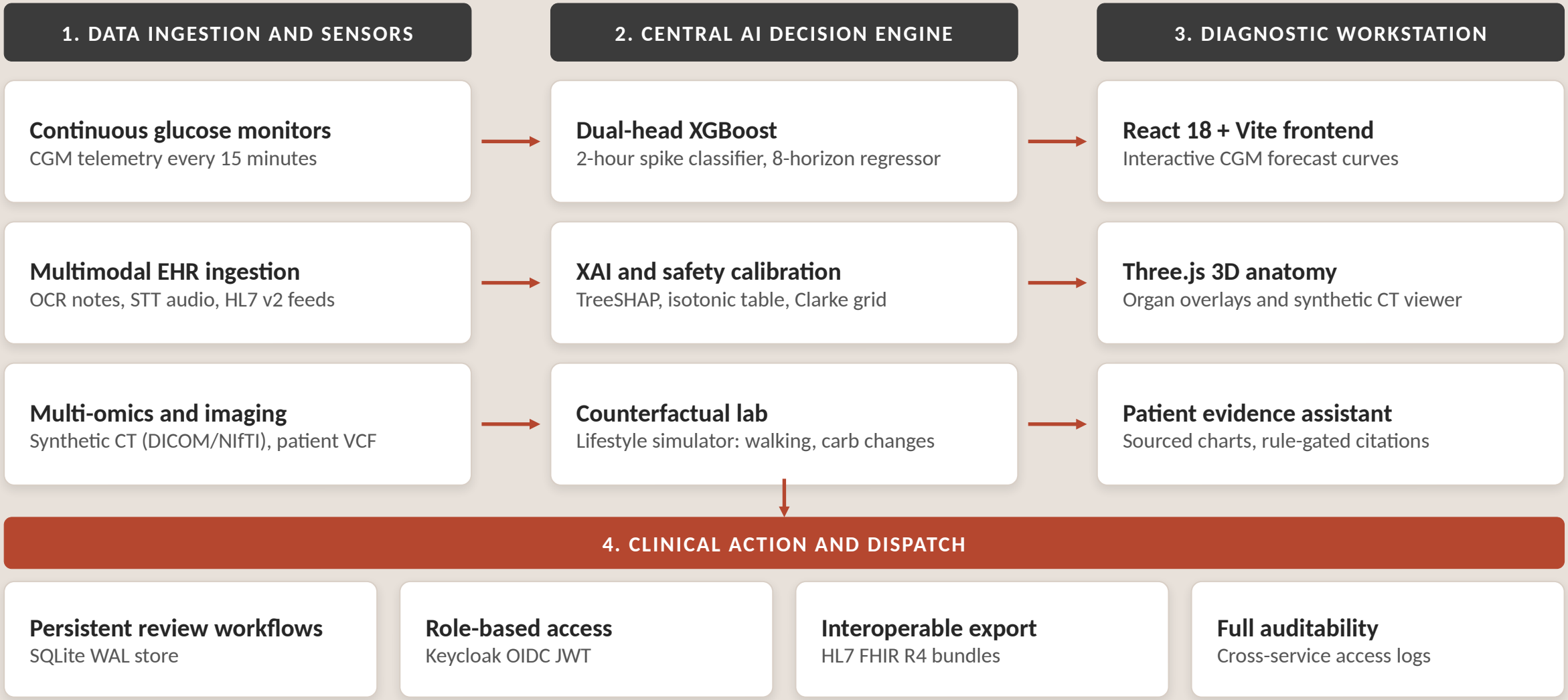
SHA-256

Deterministic serialization
Native XGBoost JSON, no pickle, checksum-verified.

Zero

Fabricated lab values
Backend owns retrieval, math and charts; the LLM cannot alter records.

System Architecture



Tech Stack

FRONTEND

React 18, **TypeScript**, **Vite**, **Three.js** (WebGL 3D organs), Recharts, Lucide Icons, scoped CSS design system

BACKEND AND SERVICES

FastAPI (Python 3.11+), Uvicorn, **Pydantic V2**, NumPy, Pandas, SciPy

MACHINE LEARNING AND XAI

XGBoost 3.2+ (native JSON), Scikit-Learn isotonic regression, **SHAP TreeExplainer**, MAPIE conformal prediction

CLINICAL SAFETY AND IMAGING

Clarke Error Grid, PyDICOM, NiBabel, Scikit-Image Marching Cubes CT-to-mesh, Trimesh

GENOMICS AND ASSISTANT

Synthetic **VCF parser** (36 T2D loci), **Groq** (llama-3.3-70b, gpt-oss-120b), OpenRouter fallback, NIDDK evidence base

STORAGE AND SECURITY

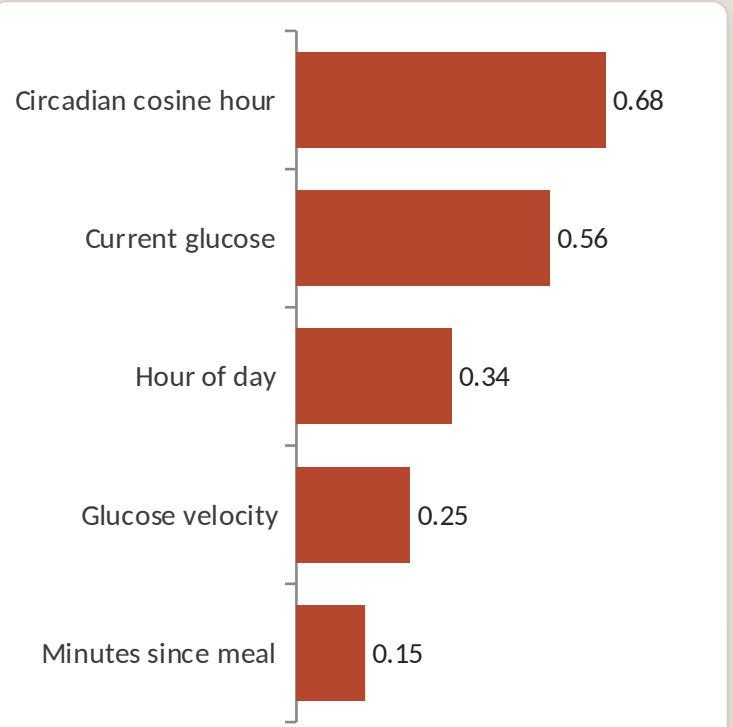
PostgreSQL 16, MinIO private storage, SQLite WAL, **Keycloak OIDC JWT**, SHA-256 fingerprinting, access audit logs

Model Performance

HEAD 1: SPIKE CLASSIFIER (180 mg/dL WITHIN 120 MIN)

Model	ROC-AUC	PR-AUC	Brier	Recall	Spec.	Prec.	FPR
AURA-Twin (calibrated)	**0.8852**	**0.6058**	**0.0907**	**57.67%**	**92.63%**	**58.87%**	**7.37%**
XGBoost (uncalibrated)	0.8861	0.6218	0.0882	67.53%	88.70%	52.23%	11.30%
Logistic regression	0.8509	0.5754	0.0953	61.35%	88.97%	50.45%	11.03%
Clinical heuristic (delta)	0.7397	0.3523	0.1726	53.45%	84.04%	38.00%	15.96%

TOP TREESHAP DRIVERS



CLINICIAN-TUNED OPERATING POINTS

Screening
Threshold 0.16
76.2% recall
16.5% false alarms

Early warning
Threshold 0.18
71.8% recall
13.2% false alarms

Low alarm
Threshold 0.22
57.7% recall
7.4% false alarms

Ward telemetry
Threshold 0.30
96.1% specificity
3.9% false alarms

HEAD 2: MULTI-HORIZON REGRESSOR AND CLINICAL SAFETY

4.82 +15 min MAE (mg/dL)

9.43 +30 min MAE (mg/dL)

15.87 +60 min MAE (mg/dL)

15.93 Mean MAE, 9.4% better than persistence

99.73% Clarke Zone A+B

Business Impact and Value

120 min
early-warning window

Under 4%
false alarms in ward mode

Rs 2-5 Lakh
ICU cost per DKA/HHS incident

40%+
lifestyle compliance boost

Hospital Enterprise SaaS

Role-gated console for **tertiary hospitals** and diabetes centers; multi-tenant pricing per bed or outpatient cohort.

Alert Fatigue Relief

Under **4%** false alarms makes ward alerts trustworthy and saves **nursing hours** every month.

ICU Cost Avoidance

DKA and HHS stays cost **Rs 2-5 Lakhs** each. A 2-hour warning allows early hydration or basal correction.

Patient Empowerment

Visual twin feedback: a brisk walk drops a **210 mg/dL** peak to **153 mg/dL**, lifting adherence.

ABDM and FHIR Ready

Native **FHIR R4** bundles plug into **ABDM**, Epic and Cerner without friction.

Scalability and Future Scope

PHASE 1 Local proof of concept → **PHASE 2 Hospital network** → **PHASE 3 National scale**

WHERE WE ARE TODAY

Working platform

Historical CGM playback, dual-head models, TreeSHAP and FHIR R4 export across **100 patient profiles**.

NEXT: CONNECT HOSPITALS

Federated learning

Train XGBoost and neural twins across hospitals **without moving raw records**, aligned with HIPAA and India's DPDP Act.

Live sensor streaming

BLE and NFC feeds from FreeStyle Libre, Dexcom G7 and indigenous biosensors.

THEN: SCALE NATIONALLY

ABDM / ABHA integration

Lifelong metabolic tracking across primary, secondary and tertiary centers.

Cardio-renal-metabolic twin

Hypertension, heart-failure ejection fraction and eGFR decline models.

Autonomous order support

Pre-filled medication, diet and nephrology orders for **one-click sign-off**.

Thank You

AURA-Twin: Precision Metabolic Intelligence for Every Patient

Questions and discussion

0.8852
ROC-AUC

99.73%
Clarke A+B

120 min
early warning

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LIVE PLATFORM AND API

Platform: localhost:5188

API docs: localhost:8004/docs

REPOSITORY

EHR Clinical Intelligence and Metabolic Digital Twin Platform